

A blockwise Perron reduction for the massless XXZ quantum transfer matrix

Candidate partial result for arXiv:2305.06679

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Status. Candidate partial result, likely valid. The global dominant-eigenvalue conjecture is *not* claimed. The result proves strict Perron dominance inside every conserved charge sector and reduces the conjecture to one cross-sector inequality.

1 Source question

Faulmann, Göhmann, and Kozłowski study the quantum transfer matrix (QTM) of the massless antiferromagnetic XXZ chain [1]. Their Conjecture 1.2, on page 11 of the source PDF, asks whether for sufficiently large Trotter number N and $T \geq a_N$ with $a_N \rightarrow 0$, the QTM has one simple eigenvalue whose modulus is strictly larger than every other eigenvalue. The same page also states their separate Conjecture 1.3 on interchange of the thermodynamic and Trotter limits.

analytic on $[0; \pi/2]$.

Finally, $\widehat{\Lambda}_{\max; \text{BA}}$ is maximal with respect to the other Eigenvalues of the quantum transfer matrix computed by the Bethe Ansatz solutions belonging to $\mathcal{B}_{\text{NLIE}}(B, \widetilde{B})$.

Here we stress that the existence and uniqueness of solutions to the mentioned non-linear integral equation and several other related non-linear problems are addressed in Theorem [2.3](#).

In order to apply our results on the rigorous analysis of solutions to the Bethe Ansatz equations to the *per se* thermodynamics of the XXZ chain, one still needs to resort to two conjectures. Most probably, their proofs will rely on techniques of analysis very different from those developed in the present work. We expect, in particular, that these conjectures hold in a much more general context and that their validity is not restricted to integrable models.

Conjecture 1.2. *For N large enough and any $T \geq a_N > 0$, for some $a_N \rightarrow 0$ as $N \rightarrow +\infty$, the quantum transfer matrix admits a dominant Eigenvalue $\widehat{\Lambda}_{\max}$ that is non-degenerate, i.e.*

$$|\widehat{\Lambda}_{\max}| > |\widehat{\Lambda}_k|, \quad \text{for any } k \in \llbracket 1; 2^{2N} - 1 \rrbracket \quad (1.36)$$

and

$$\dim\{\ker[\mathfrak{t}_q(0) - \widehat{\Lambda}_{\max}]\} = 1. \quad (1.37)$$

The second conjecture pertains to the commutativity of the thermodynamic and Trotter limits for any $T > 0$.

Conjecture 1.3. *For any $T > 0$, the thermodynamic $L \rightarrow +\infty$ and Trotter $N \rightarrow +\infty$ limits commute,*

$$\lim_{L \rightarrow +\infty} \lim_{N \rightarrow +\infty} \left\{ \frac{1}{L} \ln \text{tr}_{\mathfrak{b}_q} [(\mathfrak{t}_q(0))^L] \right\} = \lim_{N \rightarrow +\infty} \lim_{L \rightarrow +\infty} \left\{ \frac{1}{L} \ln \text{tr}_{\mathfrak{b}_q} [(\mathfrak{t}_q(0))^L] \right\}. \quad (1.38)$$

We stress that both conjectures were proven, for T large enough, in [\[39\]](#).

There are several interesting implications one can draw from these conjectures. First of all,

Figure 1: The source conjectures on page 11 of arXiv:2305.06679.

The source proves maximality only inside the class of Bethe eigenvalues it constructs at low temperature. The two global conjectures were previously proved at sufficiently high temperature [2].

2 Setup

For real p, q write

$$R(p, q) = \begin{pmatrix} p & 0 & 0 & 0 \\ 0 & q & 1 & 0 \\ 0 & 1 & q & 0 \\ 0 & 0 & 0 & p \end{pmatrix}.$$

The source uses anisotropy $\eta = -i\zeta$, with $0 < \zeta < \pi$, and

$$\aleph = -\frac{iJ \sin \zeta}{T}.$$

At $\xi = 0$ the R-matrix argument is $-\aleph/N = iu$, where

$$u = \frac{J \sin \zeta}{NT}.$$

Consequently, direct substitution into the source normalization gives

$$R(iu) = R(a, -d), \quad a = \frac{\sin(\zeta - u)}{\sin \zeta}, \quad d = \frac{\sin u}{\sin \zeta}. \quad (2.1)$$

When $0 < u < \zeta$, both a and d are strictly positive.

Let $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z = \text{diag}(1, -1)$. The QTM acts on $\mathcal{H}_q = (\mathbb{C}^2)^{\otimes 2N}$. Denote by

$$S = \prod_{k=1}^N X_{2k}, \quad P = \prod_{j=1}^{2N} Z_j$$

the even-site spin flip and total parity, respectively. Let $\mathcal{H}_m$ be the span of binary words with exactly m ones. Thus P is the scalar $(-1)^m$ on $\mathcal{H}_m$.

Define V by the same auxiliary trace as the conjugated QTM, but replace the odd local tensor by $R(a, d)$ and the even local tensor by $R(d, a)$, retaining the positive auxiliary field factor $D_0 = \exp\{h\sigma_0^z/(2T)\}$. Explicitly, with the same ordering convention as the source,

$$V = \text{tr}_0 \left[\prod_{k=N}^1 R_{0,2k}(d, a) R_{0,2k-1}(a, d) D_0 \right]. \quad (2.2)$$

This is a staggered six-vertex transfer matrix with strictly positive local Boltzmann weights.

3 Proof intuition

The negative local weight does not cause arbitrary cancellation. After flipping the even Trotter spins, two elementary Pauli- Z identities move all signs through the auxiliary product. They telescope, leaving only total quantum parity. Hence each conserved sector is a scalar sign times a nonnegative six-vertex block.

That block is not entrywise positive, because the ice rule preserves particle number, but it is primitive. A single transfer row can implement an adjacent swap $10 \leftrightarrow 01$ with two positive

c -vertices; adjacent swaps connect all words with a fixed number of ones. A no-swap row is a positive self-loop. Perron–Frobenius therefore settles every fixed sector completely. What remains is not a sign or within-sector degeneracy problem: it is the strict comparison of the $2N + 1$ sector Perron roots.

4 Blockwise Perron theorem

Theorem 1 (signed primitive-sector reduction). *Assume $J, T > 0$, $0 < \zeta < \pi$, and $0 < u < \zeta$. For the finite-Trotter QTM $t_q(0)$ of [1], the following hold.*

1. *There is the exact identity*

$$St_q(0)S = (-1)^N PV. \quad (4.1)$$

In particular, V preserves every $\mathcal{H}_m$, and the restriction of the QTM to the corresponding transformed sector is $(-1)^{N+m}V_m$, where $V_m = V|_{\mathcal{H}_m}$.

2. *Every V_m is primitive. If $\rho_m > 0$ is its Perron root, then $(-1)^{N+m}\rho_m$ is a simple QTM eigenvalue in that sector and every other eigenvalue λ in the sector satisfies $|\lambda| < \rho_m$.*
3. *The sector roots obey $\rho_m = \rho_{2N-m}$. Therefore the QTM has a unique globally dominant eigenvalue if and only if*

$$\rho_N > \rho_m \quad (0 \leq m \leq 2N, m \neq N). \quad (4.2)$$

The hypothesis holds throughout the explicit conjectural threshold regime

$$T \geq a_N := \frac{2J \sin \zeta}{N\zeta}, \quad (4.3)$$

and $a_N \rightarrow 0$.

Proof. The source monodromy alternates $R_{0,2k-1}(iu)$ with the partial transpose $R_{2k,0}^{t_{2k}}(iu)$. Direct multiplication gives

$$(X \otimes I)R(a, -d)^{t_1}(X \otimes I) = R(-d, a). \quad (4.4)$$

Thus conjugation by S turns the odd/even local tensors into $R(a, -d)$ and $R(-d, a)$.

In auxiliary–quantum tensor order, the two elementary sign identities are

$$R(a, -d) = (I \otimes Z)R(a, d)(Z \otimes I), \quad (4.5)$$

$$R(-d, a) = -(Z \otimes Z)R(d, a). \quad (4.6)$$

Insert these identities into each neighboring even–odd pair in the ordered monodromy. The auxiliary Z_0 factors cancel between consecutive pairs. Quantum Z_j factors on distinct sites commute past the other local pairs, leaving $(-1)^N P$ outside the auxiliary trace. The two surviving endpoint Z_0 factors cancel by cyclicity of the trace and because Z_0 commutes with D_0 . This proves (4.1). Ice-rule conservation gives the decomposition $V = \bigoplus_m V_m$, and $P|_{\mathcal{H}_m} = (-1)^m I$.

All six allowed local weights of V are positive. Fix m . First, $(V_m)_{x,x} > 0$ for every binary word x with m ones: keep the auxiliary arrow constant around the row and choose straight vertices. Second, whenever y is obtained from x by an adjacent swap $10 \mapsto 01$, choose the auxiliary arrow so that it exchanges with the quantum arrow at the two adjacent sites and is constant elsewhere. This produces two c -vertices and positive straight vertices, hence $(V_m)_{y,x} > 0$. The reverse swap is

obtained in the same way. Adjacent swaps on the cyclic word connect all m -particle configurations. Therefore V_m is irreducible; its positive diagonal makes it aperiodic, hence primitive. The sectorwise spectral claims now follow from the Perron–Frobenius theorem [3, Chapter 8].

For completeness, define the permutation

$$U|x_1x_2\cdots x_{2N}\rangle = |\bar{x}_{2N}\bar{x}_{2N-1}\cdots\bar{x}_1\rangle.$$

A direct matrix-element substitution in (2.2) reverses the auxiliary loop, exchanges the odd and even weights, and complements the quantum arrows; each of the six local weights and the auxiliary twist is unchanged. Hence $UVU = V$. Since $U\mathcal{H}_m = \mathcal{H}_{2N-m}$, the blocks V_m and V_{2N-m} are permutation-similar and have equal Perron roots.

If the global maximum occurred away from $m = N$, this symmetry would give it in two different sectors. Consequently global simplicity and strict modulus dominance are equivalent to (4.2). Finally, (4.3) implies $u \leq \zeta/2$, so $a, d > 0$, and clearly $a_N \rightarrow 0$. $\square$

5 Verification and failed upgrade

The included script constructs the QTM directly from the displayed monodromy, including tensor embeddings and the partial transpose. At $(\zeta, T, h, J) = (1.1, 1, 0.7, 1)$, the maximum entrywise residual in (4.1) is $0, 2.23 \cdot 10^{-16}, 4.45 \cdot 10^{-16}$, and $5.56 \cdot 10^{-16}$ for $N = 1, 2, 3, 4$, respectively. A wider grid with $N \leq 3$ confirms the predicted block signs and sectorwise gaps. These finite computations check the implementation; they are not the proof.

One plausible completion was a positive sector-raising map U_m satisfying $V_{m+1}U_m \geq U_mV_m$ entrywise. Such an inequality would imply $\rho_{m+1} \geq \rho_m$ by the Collatz–Wielandt principle. A linear-program search over all positive inclusion-supported maps is infeasible already for $N = 3, m = 2 \rightarrow 3$. Thus the simplest local raising argument does not prove (4.2).

6 Limitations and novelty check

The theorem does not prove (4.2), Conjecture 1.3, or the paper’s separate Bethe-root classification conjecture. It should therefore be reviewed only as a partial reduction. The strongest remaining route appears to require a strict finite-size comparison of Perron roots for an alternating inhomogeneous six-vertex transfer matrix, or an equivalent surface-tension inequality.

The bounded search included the source paper, the rigorous high-temperature paper [2], exact-title and exact-conjecture searches, and searches for QTM positivity, Perron–Frobenius blocks, staggered signs, and inhomogeneous six-vertex dominant sectors. It found homogeneous six-vertex sectorwise Perron results but no source stating (4.1)–(4.2) for this QTM. Novelty confidence is moderate: the crossing/sign calculation may be familiar to specialists even though no matching rigorous statement was located.

Human-review focus. Check the tensor-leg order in (4.4), the telescoping in (4.5)–(4.6), and the positive realization of adjacent swaps, including the periodic boundary. Most importantly, do not read the packet as proving the unproved cross-sector inequality (4.2).

References

- [1] S. Faulmann, F. Göhmann, and K. K. Kozłowski, *Low-temperature spectrum of the quantum transfer matrix of the XXZ chain in the massless regime*, arXiv:2305.06679.

- [2] F. Göhmann, S. Gomanee, K. K. Kozłowski, and J. Suzuki, *Thermodynamics of the spin-1/2 Heisenberg–Ising chain at high temperatures: a rigorous approach*, Comm. Math. Phys. **377** (2020), 623–673; arXiv:1811.12020.
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2012.